



Shared-Private Subspace Decomposition for Multi-Task Lung Cancer Classification in CT

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Background

N stage (LNM), M stage (DM), and EGFR mutation are *partially correlated*, yet each carries independent prognostic value; discordant combinations such as M0N1 have distinct therapeutic implications.

- **Hard sharing conflict.** Dominant nodal features suppress subtler distant-metastasis cues.
- **Shared Representation conflict.** Shared-private latent-space methods is not tailored to lung cancer CT, where shared biology and discordant-case evidence must coexist in multi-task learning.

Contributions

1 Structured Orthogonal Decoupling (SOD).

Warmup covariance initialization keeps the basis exactly orthonormal under small-batch 3D CT training.

2 Information-Inspired Subspace Regularization (ISR).

Private-only supervision, cross-covariance minimization, and VIB compression keep private subspaces discriminative and non-redundant.

Datasets & Setup

6,374

Huaxi · internal

1,146

Shengjing · external

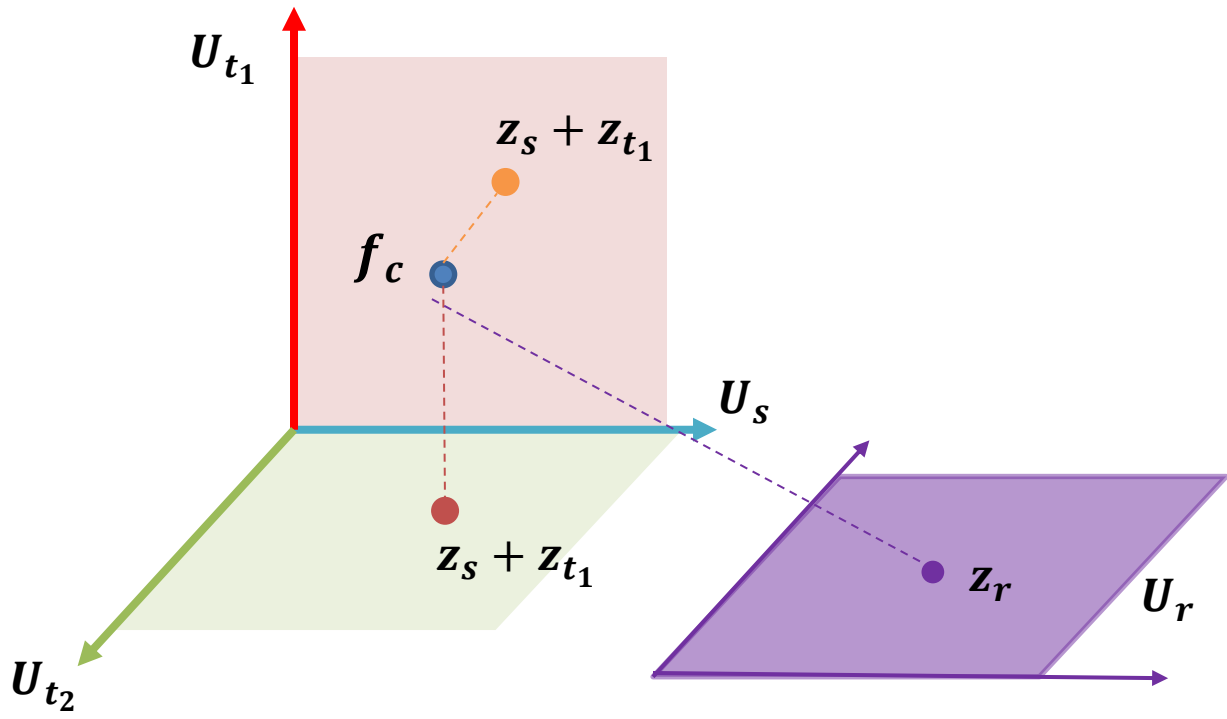
Datasets: 3D CT; lung VOI cropping, resampled to 224×224×48, Z-score normalized; Huaxi train/val/test = 7:1:2, three binary labels (LNM, DM, EGFR); Shengjing provides LNM and DM only (28.2% / 18.0% positive) — a domain-shift test.

Implementation: 3D DenseNet (16 filters), d = 128; private k = 24/task, shared k_s = 8, residual k_r = 48; 40 epochs, batch 16, lr 5×10⁻⁴ cosine; all baselines share the same backbone and protocol.

Core idea

A compact shared space plus larger private spaces gives the best trade-off.

The largest gains appear in mAP and EMR — better joint-label decisions, not only better average discrimination.)

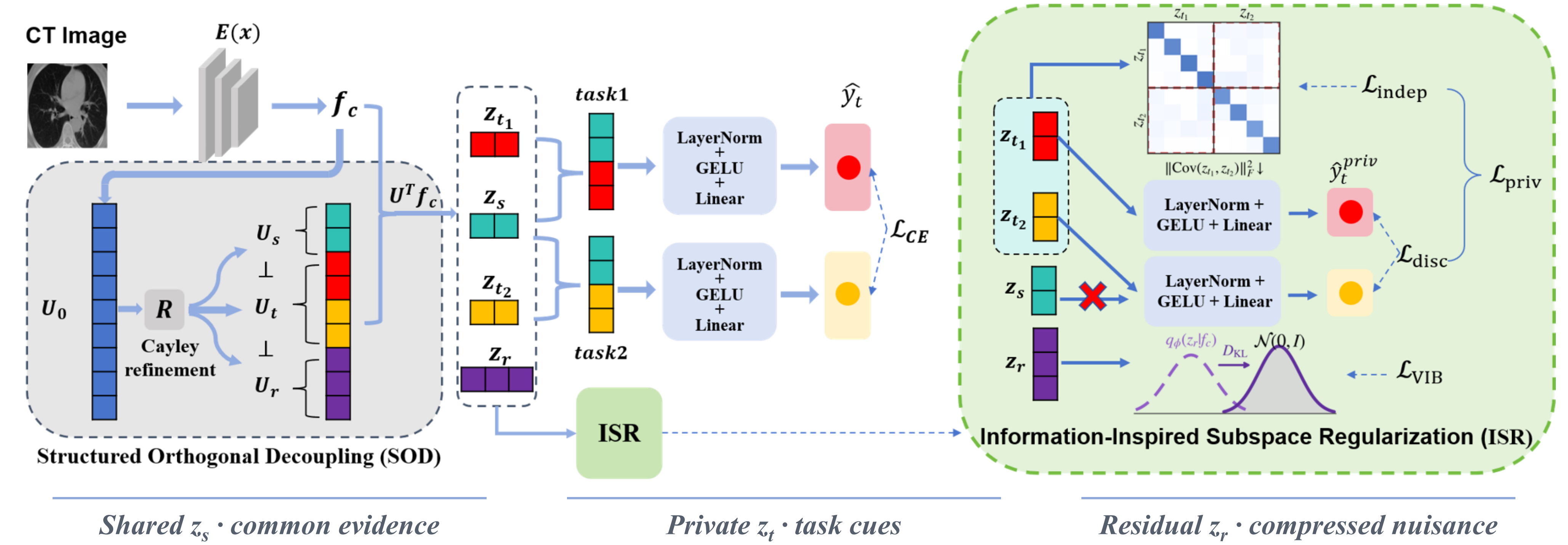


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Framework



Shared z_s · common evidence

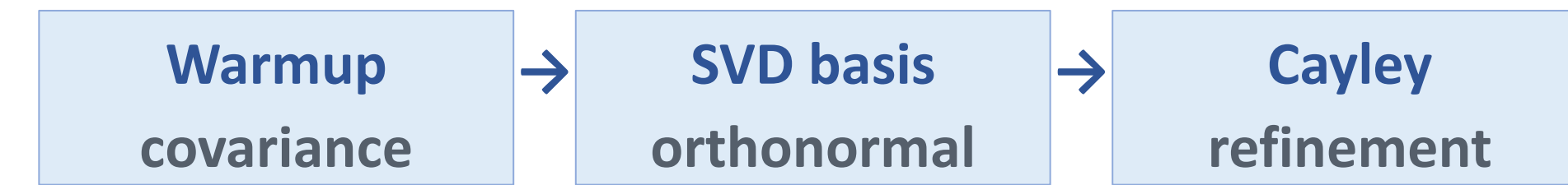
Private z_t · task cues

Residual z_r · compressed nuisance

Structured Orthogonal Decoupling

- Feature $f = E(x)$ is split into mutually orthogonal bases: *shared* U_s (common evidence), *private* U_t per task, *irrelevant* U_r (nuisance); $\text{span}(U_o) \perp \text{span}(U_b)$.
- U_o from SVD of warmup feature covariance, refined via the Cayley transform, ensuring orthogonality holds throughout training.

Two-stage basis construction



Information-Inspired Regularization

The Optimization Objective:

$$\max \sum_t I(z_s; z_t; y_t) + \alpha \sum_t I(z_t; y_t) - \beta \sum_{a < b} I(z_a; z_b) - \gamma I(z_r; f_d)$$

- L_{CE} — cross-entropy on $[z_s; z_t]$ keeps predictions label-informative.
- $L_{priv} = L_{disc} + \eta \cdot L_{indep}$ — private-only branch keeps z_t discriminative; $\|Cov(z_a, z_b)\|_F^2$ removes redundancy.
- L_{VIB} — KL of $q_\phi(z_r|f_d)$ to $N(0, I)$ compresses residual variation.

$$\text{Final objective: } L = L_{CE} + 0.1 \cdot L_{priv} + 0.1 \cdot L_{VIB} (\eta = 0.2)$$

$$d = 128 = \text{shared}(8) + \text{private}(24 \times 3) + \text{residual}(48)$$

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Internal Test

Method	mACC	mAUC	mAP	EMR	M0N1 R	M0N1 F1
Single-Task	0.7202	0.7841	0.7813	0.3834	0.3687	0.3931
Multi-Task	0.7346	0.7725	0.7657	0.3529	0.3180	0.3239
GradNorm	0.7208	0.7934	0.7820	0.3771	0.2662	0.3083
CPCL	0.7346	0.8064	0.7772	0.3682	0.2982	0.3103
PLE	0.7359	0.8033	0.7816	0.3911	0.2637	0.3178
CIPL	0.7146	0.7983	0.7826	0.3694	0.3226	0.2578
Ours	0.7394	0.8110	0.7998	0.4111	0.3463	0.3708

- **Negative transfer is measurable:** MTL and GradNorm fall below Single-Task on EMR.
- **On the discordant M0N1 subgroup:** improve recall and F1 over all multi-task baselines.
- **Gains over the strongest baseline.**

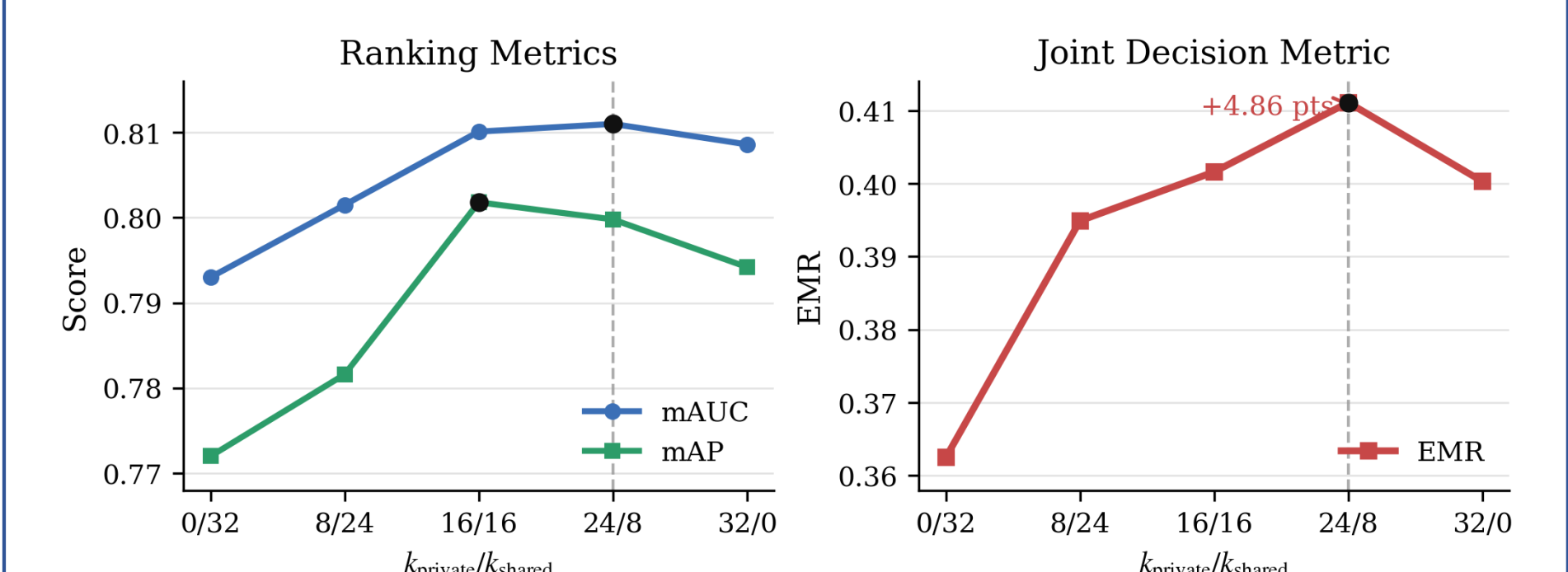
External Validation

Shengjing (external)	mACC	mAUC	EMR
MTL	0.7295	0.7294	0.596
CIPL	0.7395	0.6847	0.5611
MTL + VIB	0.7312	0.7420	0.5503
Ours w/o VIB	0.7565	0.7385	0.6117
Ours	0.7648	0.7722	0.6335

- **Best under domain shift** (different center, various reconstruction kernels; LNM / DM only).
- **+0.0353 mACC · +0.0428 mAUC · +0.0375 EMR**

Ablation & Capacity Allocation

Ablation	mACC	mAUC	mAP	EMR
MTL (baseline)	0.7346	0.7725	0.7657	0.3529
+ SOD	0.7236	0.7968	0.7728	0.3787
+ Lpriv	0.7362	0.8104	0.7976	0.4137
Full (SOD + ISR)	0.7394	0.8110	0.7998	0.4111



- EMR peaks at $k_{\text{private}} = 24$, $k_{\text{shared}} = 8$; removing the shared subspace degrades performance.